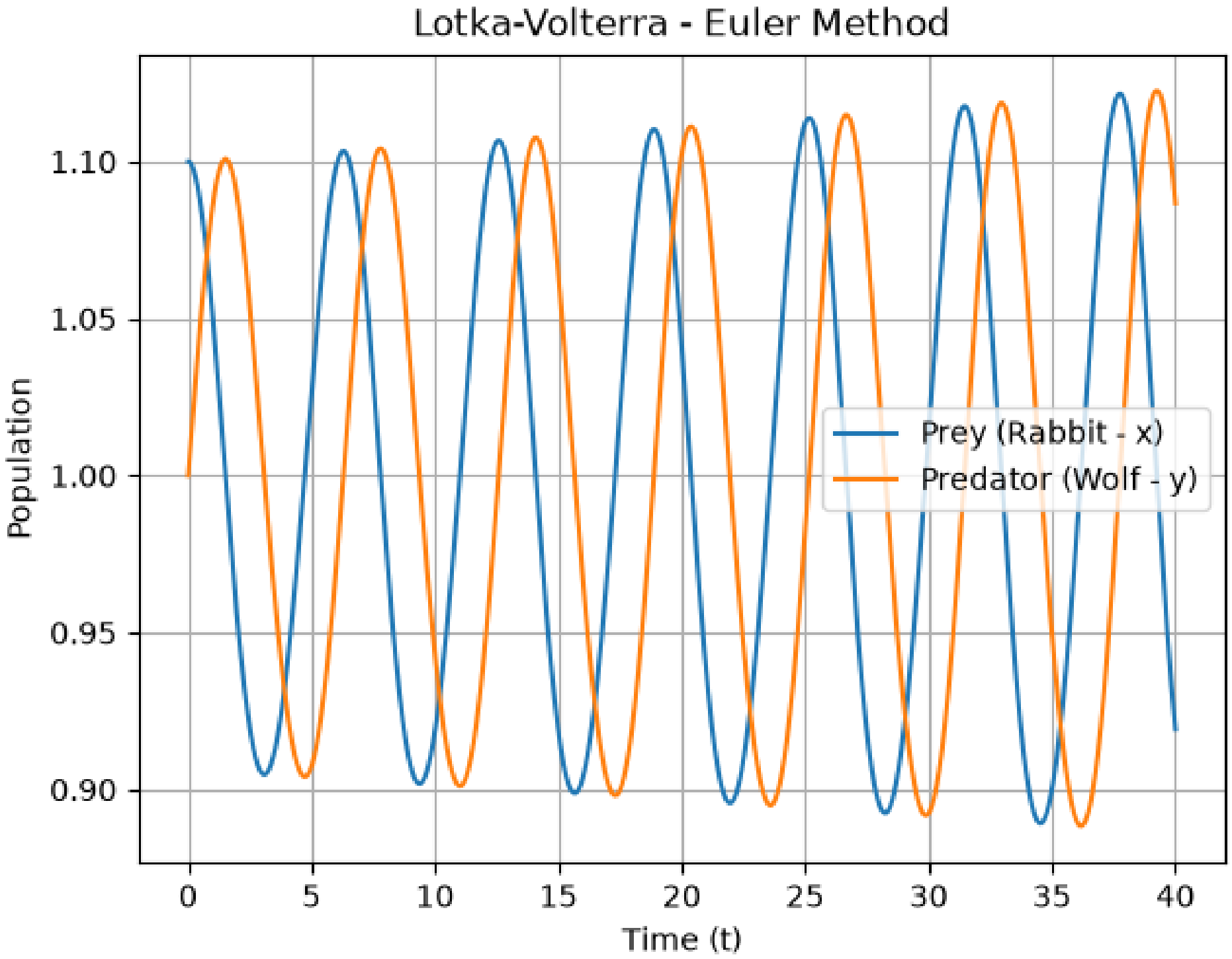
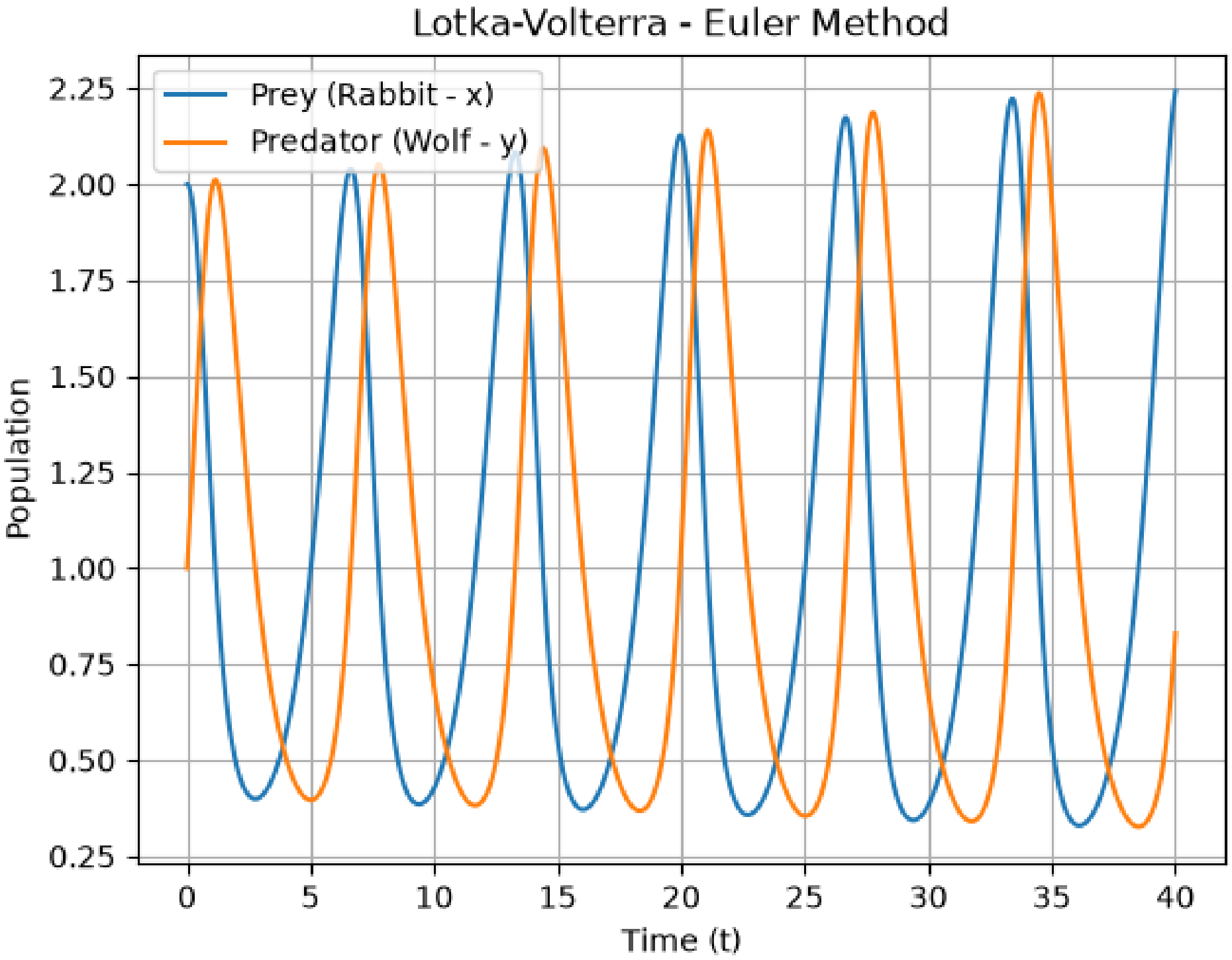
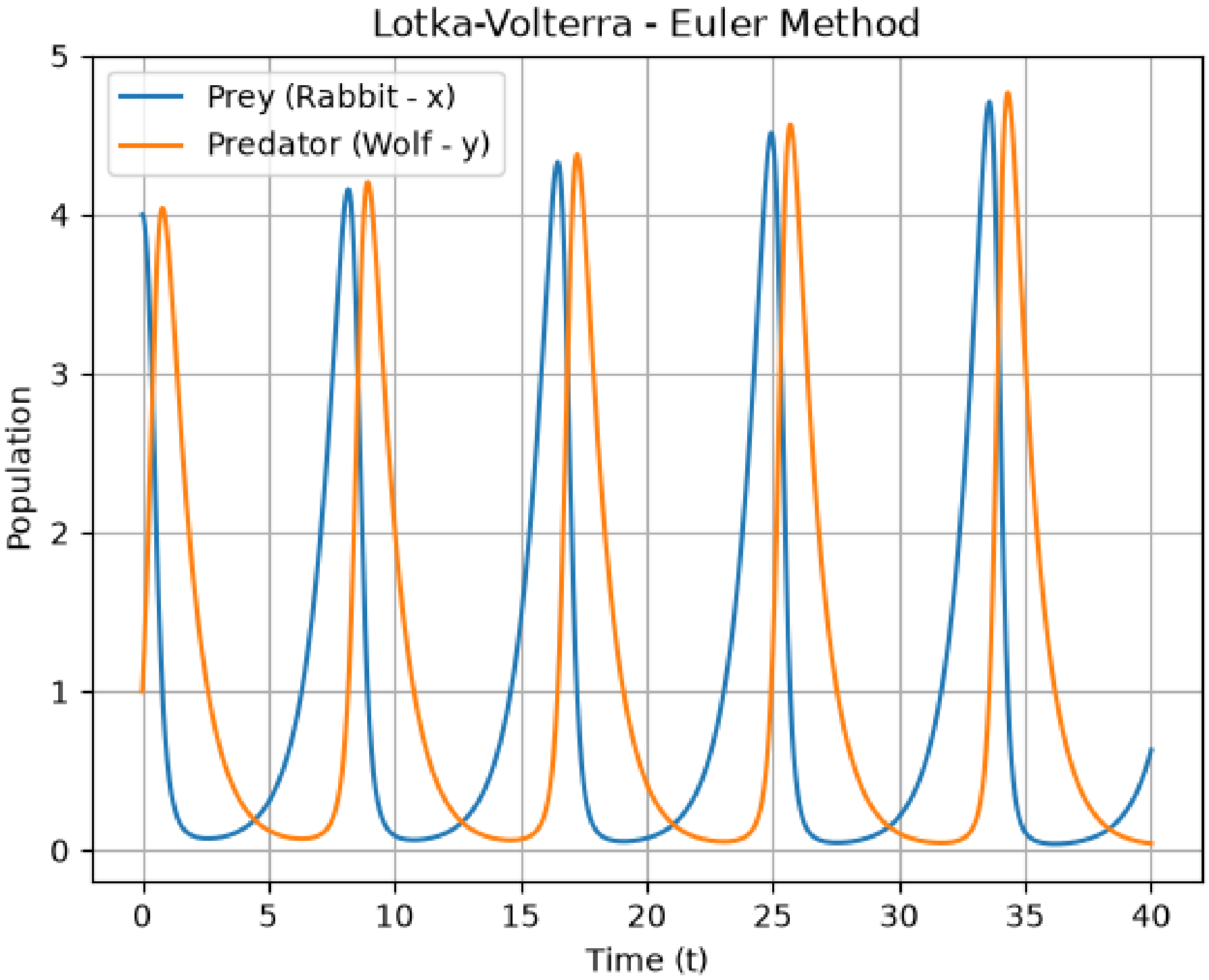
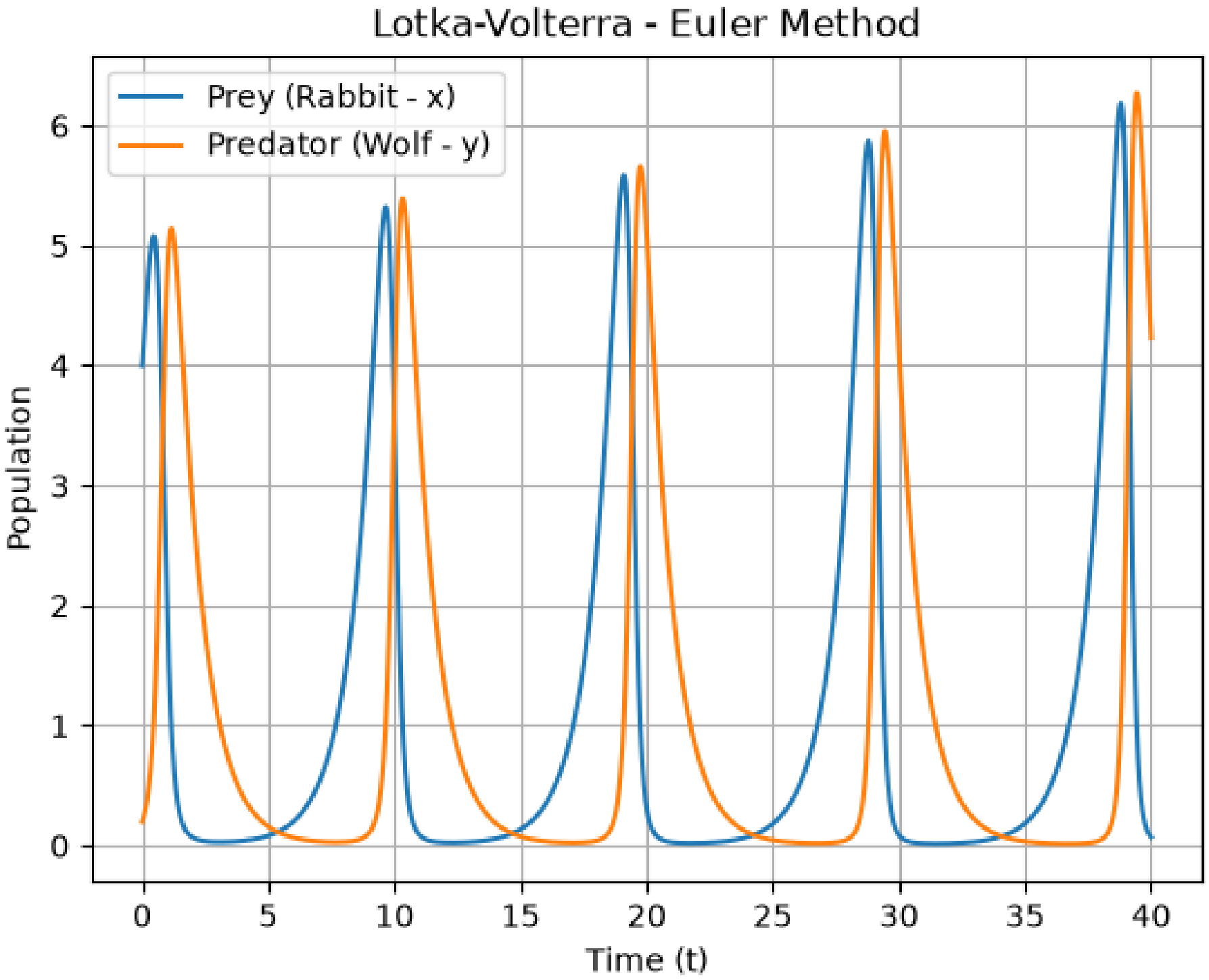
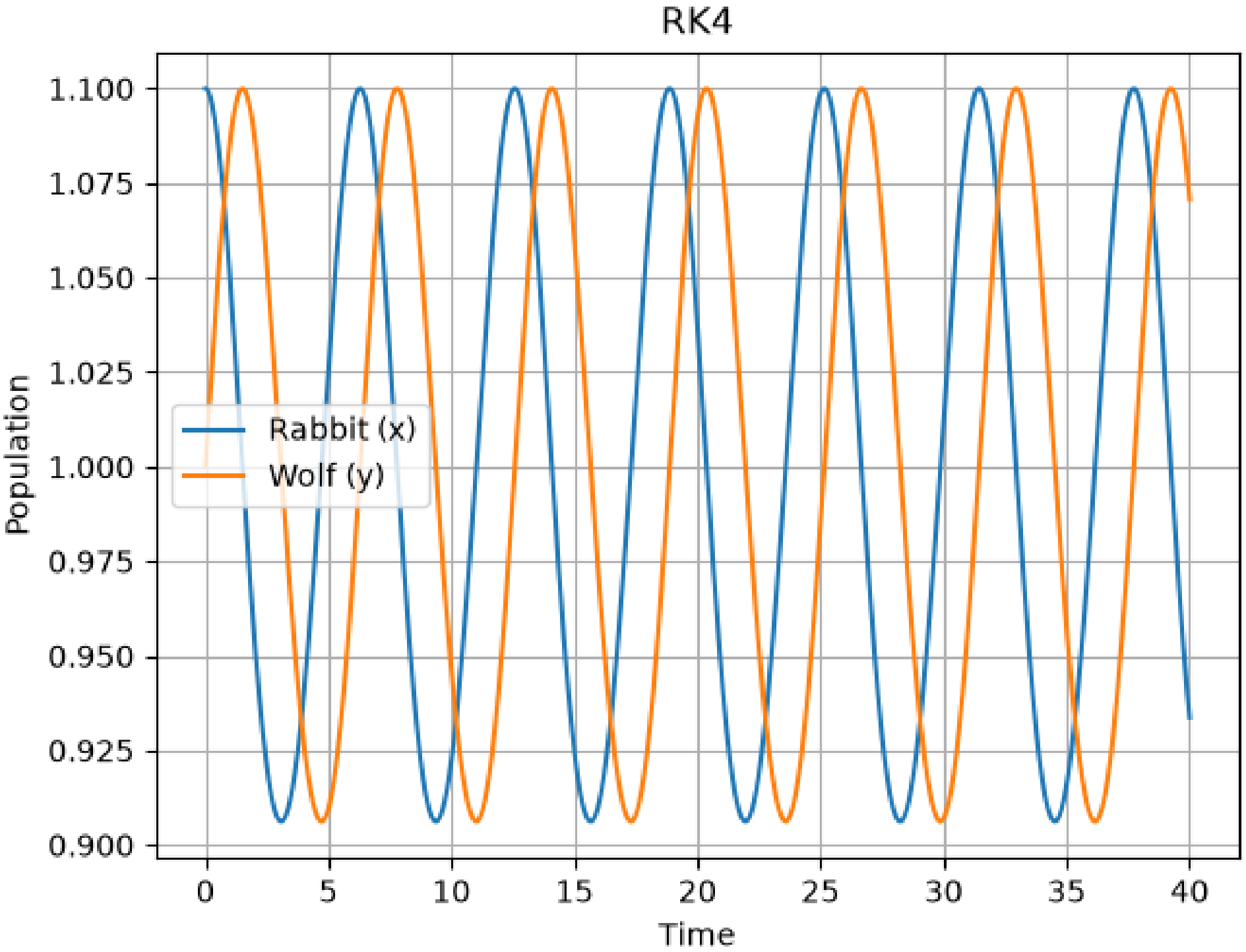
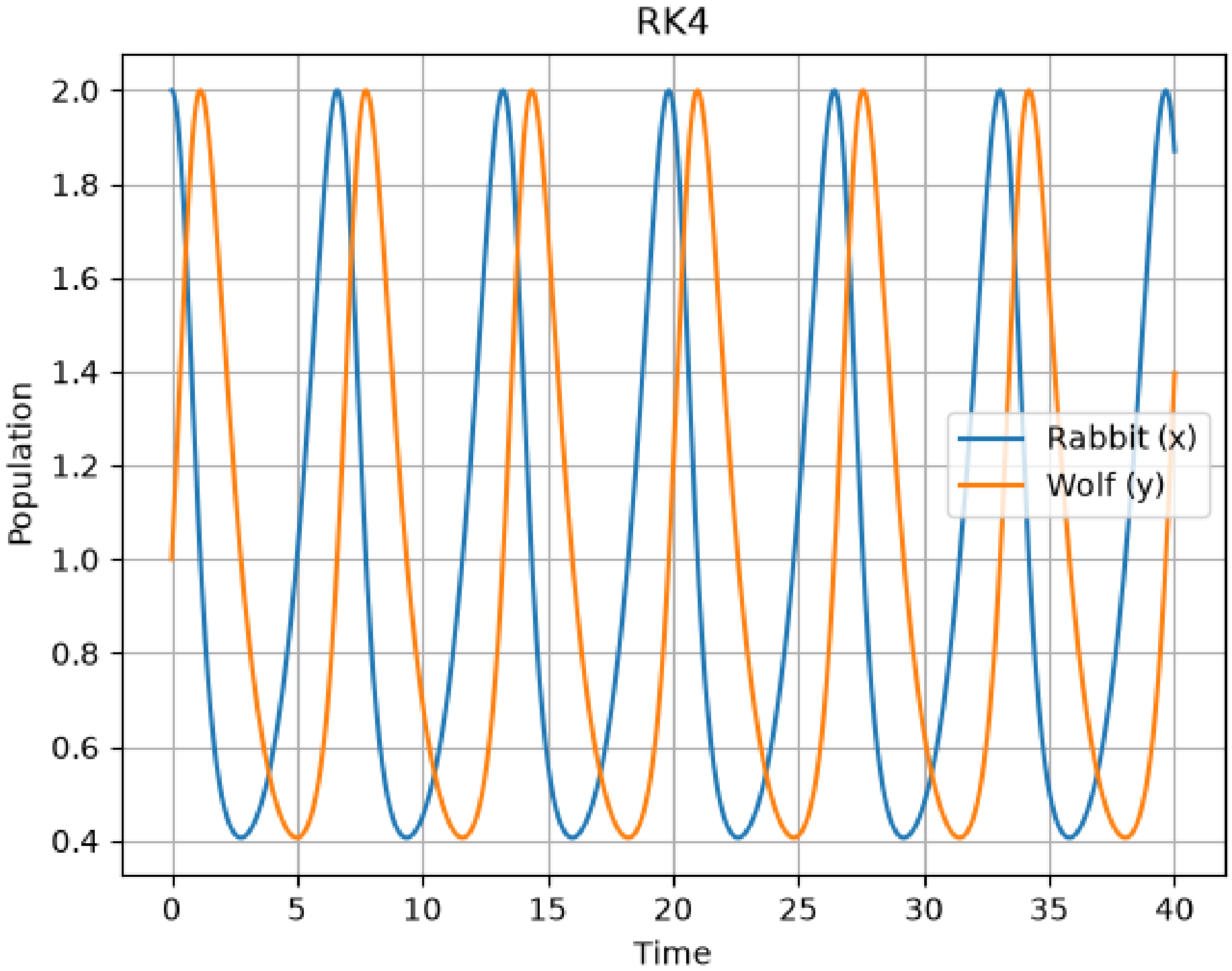

Lotka-Volterra - Euler Method
Population
1.10
1.05
1.00
0.95
0.90
0
5
10
15
20
25
30
35
40
Time (t)
Prey (Rabbit - x)
Predator (Wolf - y)

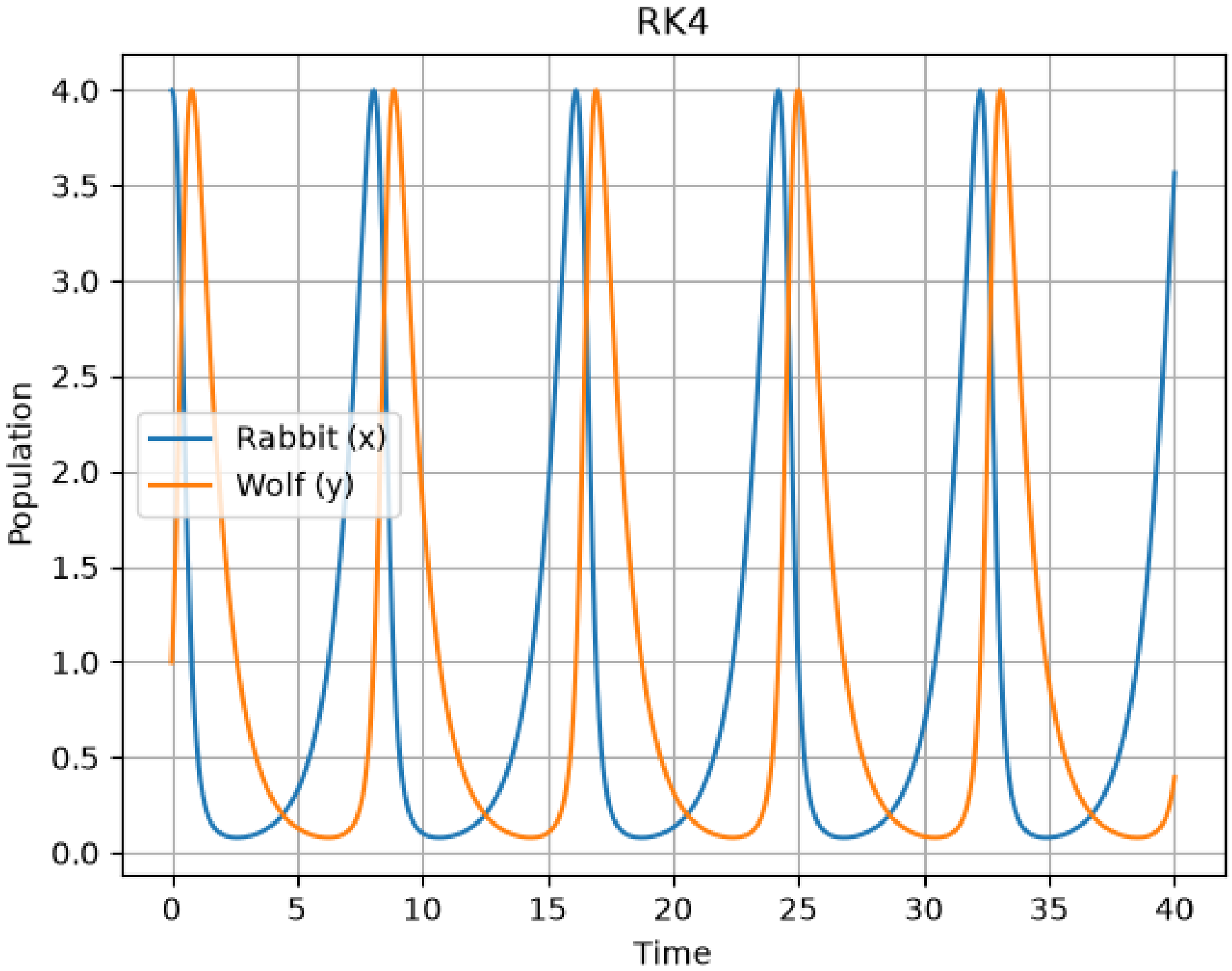

Lotka-Volterra - Euler Method
Prey (Rabbit - x)
Predator (Wolf - y)
Population
Time (t)
2.25
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0
5
10
15
20
25
30
35
40

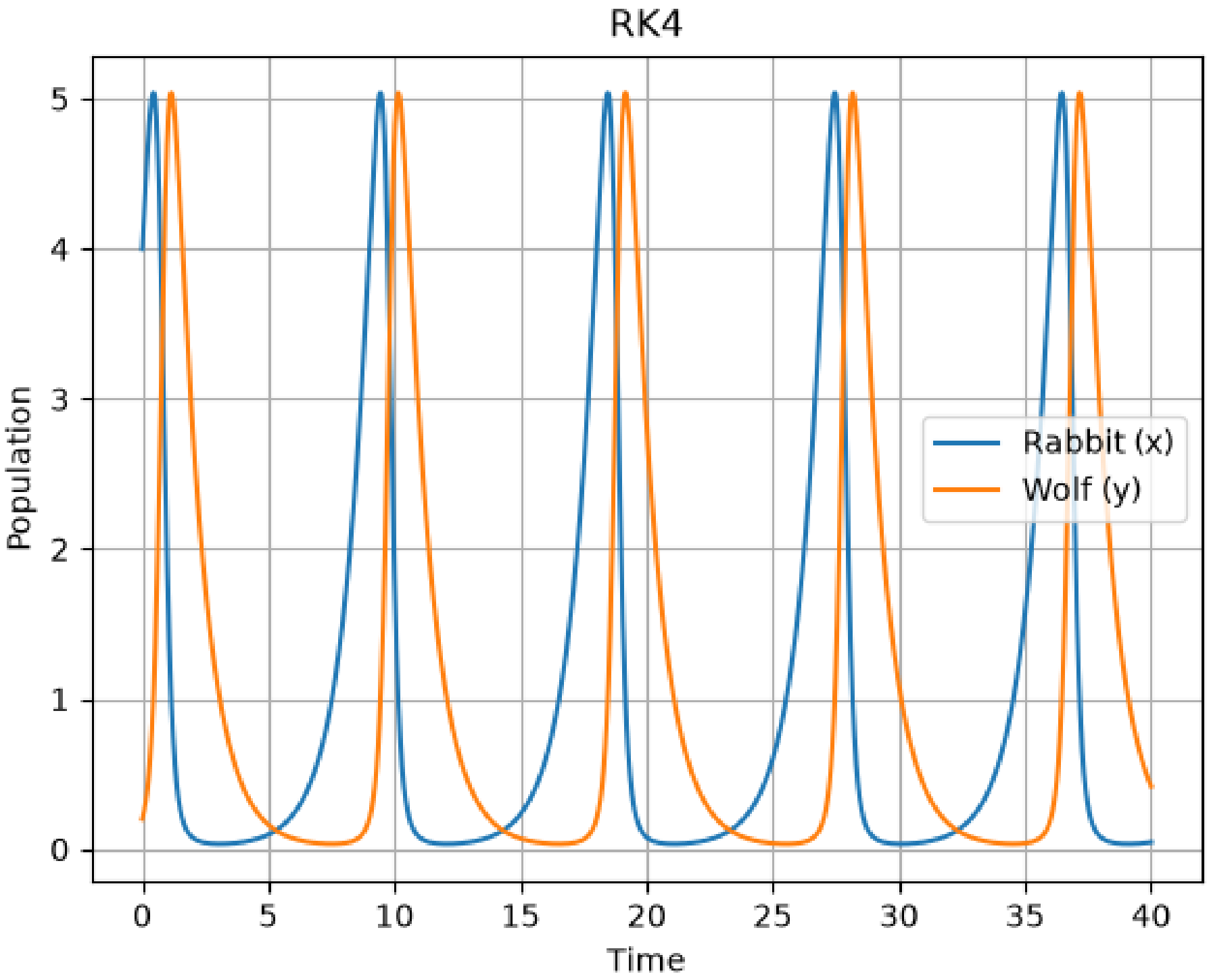

Lotka-Volterra - Euler Method
Prey (Rabbit - x)
Predator (Wolf - y)
Population
Time (t)
0
1
2
3
4
5
0
5
10
15
20
25
30
35
40

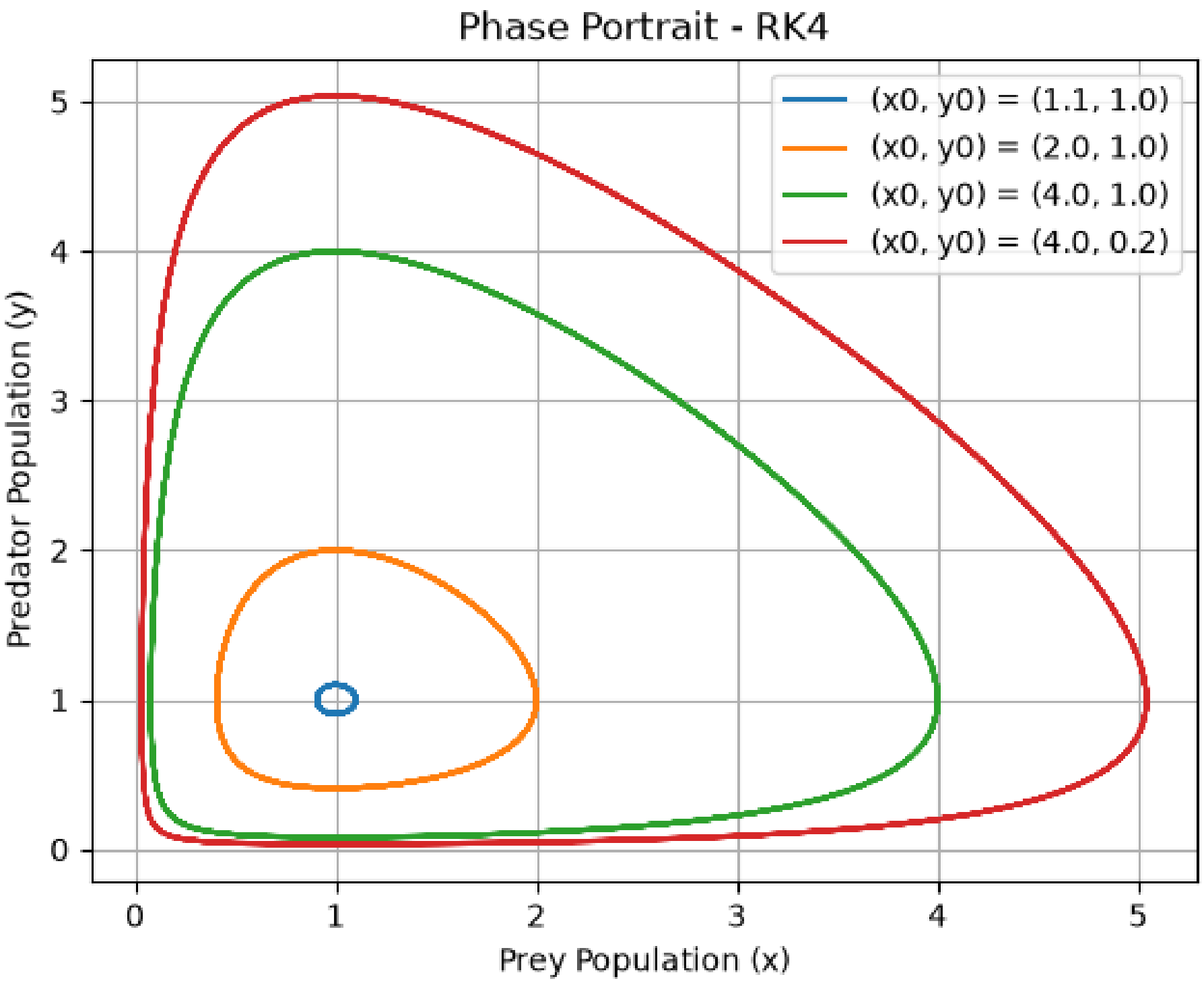

Lotka-Volterra - Euler Method
Prey (Rabbit - x)
Predator (Wolf - y)
Population
Time (t)
0
1
2
3
4
5
6
0
5
10
15
20
25
30
35
40


RK4
Population
Time
Rabbit (x)
Wolf (y)
1.100
1.075
1.050
1.025
1.000
0.975
0.950
0.925
0.900
0
5
10
15
20
25
30
35
40


RK4
Population
Time
Rabbit (x)
Wolf (y)
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0
5
10
15
20
25
30
35
40


RK4
Rabbit (x)
Wolf (y)
Population
Time
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
5
10
15
20
25
30
35
40


RK4
Population
Time
Rabbit (x)
Wolf (y)
0
1
2
3
4
5
0
5
10
15
20
25
30
35
40


Phase Portrait - RK4
(x0, y0) = (1.1, 1.0)
(x0, y0) = (2.0, 1.0)
(x0, y0) = (4.0, 1.0)
(x0, y0) = (4.0, 0.2)
Predator Population (y)
Prey Population (x)
0
1
2
3
4
5